Power Output Manipulation from Below to Above the Gas Exchange Threshold Results in Exacerbated Performance Fatigability

CALLUM G. BROWNSTEIN[1], FREDERIC SABATER PASTOR[1], JOSÉ MIRA[1], JUAN M. MURIAS[2], and GUILLAUME Y. MILLET[1,3]

[1]Inter-university Laboratory of Human Movement Biology, UJM-Saint-Etienne, University of Lyon, EA 7424, F-42023, Saint-Etienne, FRANCE; [2]Faculty of Kinesiology, University of Calgary, Calgary, Alberta, CANADA; and [3]Institut Universitaire de France (IUF), Paris, FRANCE

ABSTRACT

BROWNSTEIN, C. G., F. S. PASTOR, J. MIRA, J. M. MURIAS, and G. Y. MILLET. Power Output Manipulation from Below to Above the Gas Exchange Threshold Results in Exacerbated Performance Fatigability. *Med. Sci. Sports Exerc.*, Vol. 54, No. 11, pp. 1947–1960, 2022. **Introduction:** Performance fatigability is substantially greater when exercising in the severe- versus heavy-intensity domain. However, the relevance of the boundary between moderate- and heavy-intensity exercise, the gas exchange threshold (GET), to performance fatigability is unclear. This study compared alterations in neuromuscular function during work-matched exercise above and below the GET. **Methods:** Seventeen male participants completed work-matched cycling for 90, 110, and 140 min at 110%, 90%, and 70% of the GET, respectively. Knee extensor isometric maximal voluntary contraction (MVC), high-frequency doublets (Db100), low- to high-frequency doublet ratio (Db10:100), and voluntary activation were measured at baseline, 25%, 50%, 75%, and 100% of task completion. During the initial baseline visit and after each constant work rate bout, ramp-incremental exercise was performed, and peak power output and oxygen uptake ($\dot{V}O_{2peak}$) were determined. **Results:** After the 70% and 90% GET trials, similar reductions in MVC (−14% ± 6% and −14% ± 8%, respectively, P = 0.175) and Db100 (−7% ± 9% and −6% ± 9%, respectively, P = 0.431) were observed. However, for a given amount of work completed, reductions in MVC (−25% ± 15%, P = 0.008) and Db100 (−12% ± 8%, P = 0.029) were up to 2.6-fold greater during the 110% than the 90% GET trial. Peak power output and $\dot{V}O_{2peak}$ during ramp-incremental exercise were reduced by 7.0% ± 11.3% and 6.5% ± 9.3%, respectively, after the 110% GET trial relative to the baseline ramp ($P \leq 0.015$), with no changes after the moderate-intensity trials ($P \geq 0.078$). **Conclusions:** The lack of difference in fatigability between the trials at 70% and 90% GET, coupled with the greater fatigability at 110% relative to 90% GET, shows that exceeding the moderate- to heavy-intensity boundary has implications for performance fatigability, while also impairing maximal exercise performance capacity. **Key Words:** CONTRACTILE FUNCTION, CORTICOSPINAL EXCITABILITY, EXERCISE INTENSITY, INTENSITY DOMAINS, MOTONEURON EXCITABILITY, NEUROMUSCULAR FUNCTION, VOLUNTARY ACTIVATION

Whole-body locomotor exercise evokes substantial increases in energy demands, requiring a coordinated, integrative response of multiple physiological systems (1). The demands imposed on these physiological systems are set in large part by the external work demand (i.e., power output or speed), which determines the rate of utilization of adenosine triphosphate (ATP) and in turn muscle metabolism. Beyond specific metabolic rates, termed thresholds, there are relatively abrupt disruptions in metabolic homeostasis, which are identifiable through shifts in the gas exchange and ventilatory responses to exercise (2). The first of these thresholds, termed the gas exchange threshold (GET; the gas exchange analogue of the lactate threshold), represents the highest metabolic rate not associated with metabolic acidosis and a loss of efficiency leading to the slow component in oxygen uptake ($\dot{V}O_{2SC}$) (3). The second threshold, termed the maximum metabolic steady state, is frequently determined through critical power (3) or the maximal lactate steady state (4) and represents the maximum metabolic rate at which ATP resynthesis can be supported almost exclusively by oxidative phosphorylation, above which no steady state in oxidative phosphorylation or substrate level phosphorylation can be attained. Respectively, these two thresholds demarcate the domains of moderate from heavy, and heavy from severe-intensity exercise.

The metabolic perturbations associated with exercise in the severe-intensity domain have important implications for performance fatigability, defined as an acute exercise-induced impairment in the force or power generating capacity of the

Address for correspondence: Callum G. Brownstein, Ph.D., Laboratoire Interuniversitaire de Biologie de la Motricité, Bâtiment IRMIS, 10 rue de la Marandière, 42270 Saint Priest en Jarez, France; E-mail: callum.brownstein@univ-st-etienne.fr.

Submitted for publication March 2022.
Accepted for publication June 2022.
Supplemental digital content is available for this article. Direct URL citations appear in the printed text and are provided in the HTML and PDF versions of this article on the journal's Web site (www.acsm-msse.org).

0195-9131/22/5411-1947/0
MEDICINE & SCIENCE IN SPORTS & EXERCISE®

DOI: 10.1249/MSS.0000000000002976

involved muscles (5), the etiology of which can be measured using neurostimulation methods (6). Specifically, the inexorable contribution of substrate level phosphorylation results in a continuous production of contractile function-impairing metabolites, such as inorganic phosphate (Pi) and hydrogen (H^+) (7,8). Indeed, Burnley et al. (9) showed that the rate of impairments in isometric maximal voluntary torque and contractile function, measured through electrically evoked responses to relaxed muscle, was four to five times faster when performing isometric exercise moderately above versus below critical torque. Consequently, even power outputs marginally above critical power or the maximum lactate steady state (10) are associated with inexorable perturbations in metabolic homeostasis and, concurrently, impairments in neuromuscular function owing primarily to disturbances at the contractile level.

Although there is strong evidence on the importance of the boundary between heavy- and severe-intensity exercise as a "performance fatigability threshold," the neuromuscular consequences of moderate- versus heavy-intensity exercise are unclear. Few studies have compared the neuromuscular responses to sub- and supra-GET exercise. While Black et al. (11) found that task failure was reached after a substantially shorter duration during heavy- versus moderate-intensity exercise (44 ± 16 vs 211 ± 57 min, respectively), vast differences were present in the exercise intensity between the two trials (87% ± 4% $\dot{V}O_{2peak}$, 231 ± 56 W for heavy-intensity exercise and 52% ± 8% $\dot{V}O_{2peak}$, 113 ± 19 W for moderate-intensity exercise). As such, although the shorter time to task failure indicates that performance fatigability was exacerbated in the heavy domain, it is unclear whether this was a result of being in a higher-intensity domain *per se* or simply a result of the much greater metabolic rate. When assessing fatigability after 8 min of moderate-, heavy-, and severe-intensity exercise, Cannon et al. (12) found that peak power during isokinetic cycling was reduced after severe- and heavy- but not moderate-intensity exercise. Although these results suggest that exceeding the GET could have implications for neuromuscular function, neither of the aforementioned studies used neurostimulation methods to determine the etiology of impaired neuromuscular function. Thus, the relevance of the boundary between moderate- and heavy-intensity exercise in regard to performance fatigability remains unclear and could have implications for neuromuscular impairments induced by endurance exercise and/or activities of daily living, particularly in patient groups whose GET occurs at low work rates (13).

There is reason to suspect that fatigability would be present during both moderate- and heavy-intensity exercise, but that fatigability would be exacerbated when exercising at the latter intensity. Although the concentration of contractile function-impairing metabolites stabilizes during moderate-intensity exercise, they are nevertheless elevated relative to rest (14,15). This is in large part due to the time required for oxidative phosphorylation to meet ATP demands at the onset of exercise, i.e., the $\dot{V}O_2$ on-kinetics. However, during heavy-intensity exercise, the loss of efficiency necessitates an increased output through both oxidative and nonoxidative energy pathways, resulting in a disproportionately higher $\dot{V}O_2$, PCr degradation, and metabolite accumulation, before the delayed attainment of a metabolic steady state (14,15). This loss of efficiency has potential implications for performance fatigability, with several studies demonstrating a relationship between the $\dot{V}O_{2SC}$ and reductions in neuromuscular function (12,16,17). Moreover, the increase in the ATP cost of force generation above the GET (18) has implications for glycogen depletion-related impairments in excitation–contraction coupling during prolonged bouts of exercise because of the disproportionally higher energy demand relative to moderate-intensity exercise (2). Accordingly, several mechanisms could contribute to impaired fatigability when exercising in the heavy versus moderate domain. However, whether this is indeed the case, and to what extent, is unknown.

The primary aim of this study was to assess performance fatigability during constant work rate, work-matched exercise above and below the GET. To assess the implications of sub- and supra-GET exercise on subsequent exercise performance, a secondary aim of this study was to assess peak power output during ramp-incremental exercise performed after the constant work rate bouts. It was hypothesized that impairments in neuromuscular function would be present during bouts of exercise at 70% and 90% GET, but that fatigability would be greater at 110% than 90% GET, with no difference between 90% and 70% GET. Moreover, we hypothesized that impairments in subsequent exercise performance would be exacerbated in response to work-matched exercise at supra- compared with sub-GET.

METHODS

Participants

The study received ethical approval from the local ethics committee and conformed to the standards set by the Declaration of Helsinki, except for registration in a database. Participants provided written informed consent to take part in the study. Seventeen healthy male participants were recruited for the study (age, 26 ± 4 yr; stature, 179.8 ± 7.8 cm; mass, 70.9 ± 8.2 kg). All participants were undertaking a regular exercise regime ranging from recreational fitness to amateur competitive sport.

Design

Participants visited the laboratory on four occasions, including one baseline visit during which a familiarization with the neuromuscular testing procedures was performed, followed by a ramp-incremental exercise test to task failure. The three subsequent visits consisted of work-matched constant work rate exercise at 110%, 90%, and 70% of the GET. A work-matched approach was deemed preferable to time-matched approach, given that any differences in fatigability using the latter approach could be the result of the greater work performed at higher intensities rather than the higher intensity *per se*. Five minutes after the three constant work rate trials, ramp-incremental exercise was performed using the same protocol as the baseline visit. The order of the three constant work rate trials was randomized. The time of day for each testing session was controlled (±2 h) to account for diurnal physiological variations. Visits to the laboratory were separated by a minimum of 1 wk and a maximum of 3 wk. Laboratory temperature was monitored during each trial and was consistent during the three constant work rate trials (25.0°C ± 1.5°C, 24.8°C ± 1.2°C, and 24.9°C ± 1.6°C at 110%, 90%, and 70% of the GET, respectively, repeated-measures ANOVA $P = 0.878$).

Experimental Protocol

Familiarization and incremental exercise test. Before the incremental exercise, participants were familiarized with the neuromuscular testing procedures, including performing maximal and submaximal isometric knee extensor contractions and receiving all forms of neurostimulation used in the protocol. Subsequently, a square wave transition followed by a ramp-incremental test to task failure was performed on a custom-built, electromagnetically braked recumbent cycle ergometer. The square wave transition was used to account for the $\dot{V}O_2$ mean response time (MRT; described below) and consisted of a modified protocol of that developed by Iannetta et al. (19). Specifically, participants cycled at 20 W for 2 min, followed by 6 min at 100 W, and 4 min at 20 W. Immediately after cycling for 4 min at 20 W, the ramp-incremental protocol began, commencing at 20 W with a 30-W·min^{-1} increment. The test was terminated once the participant's self-selected cadence decreased by 10 rpm, or they voluntarily disengaged from the task despite strong verbal encouragement, whichever came sooner. During the square wave and ramp-incremental protocols, breath-by-breath pulmonary gas exchange and ventilation was measured (MetaMax 3B, Cortex Medical, Leipzig, Germany) as well as HR (Incorpus, Becare, Renens, Switzerland). Outcome variables from the ramp-incremental test included $\dot{V}O_{2peak}$ (mL·kg^{-1}·min^{-1} and L·min^{-1}), peak power output (W), peak HR (bpm), and the $\dot{V}O_2$ (mL·kg^{-1}·min^{-1} and %$\dot{V}O_{2peak}$) and power output (W) associated with the GET, with the latter calculated taking into account the MRT (described below).

Constant work rate trials. The protocol for the constant work rate trials is depicted in Figure 1. Before the first constant work rate trial, participants recorded their diet the day before and the day of the constant work rate trials and were asked to replicate their diet over the same periods before subsequent visits. Participants were asked to abstain from strenuous exercise in the 24 h before each trial. The constant work rate trials consisted of worked-matched cycling at 110%, 90%, and 70% of the power output associated with the GET. The duration of cycling at the target power outputs during these trials was 90, 110, and 140 min, respectively, with 2 min of baseline cycling at 20 W performed immediately before each trial. Cycling was performed on the same custom-built recumbent ergometer as the ramp-incremental exercise test, which permits the immediate assessment of knee extensor neuromuscular function after cycling (20). During each trial, neuromuscular assessments (described below) were performed at baseline, 25%, 50%, 75%, and 100% of task completion. These assessments began within 5 s of cycling cessation and took approximately 50 s to complete. Overall, the pause in constant work rate cycling during which neuromuscular assessments took place lasted 3 min, with participants cycling at 20 W for the remainder of this duration once the neuromuscular assessment was complete. Participants were permitted to drink water *ad libitum*. During the first constant work rate trial, participants were permitted to self-select their cadence (60–80 rpm) and were required to maintain this cadence for the rest of the trial and during the subsequent trials. Blood lactate was measured after 1 min of baseline cycling at 20 W and 1 min before each neuromuscular assessment throughout the constant work rate trials (Fig. 1). To confirm the presence or absence of a $\dot{V}O_{2SC}$, gas exchange measurements were taken throughout the entirety of the initial 25% of the task. Thereafter, gas exchange measures were taken for a period of 3 min, terminating 1 min before each subsequent neuromuscular assessment (i.e., those at 50%, 75%, and 100% of task completion; Fig. 1). Outcome variables during the constant work rate trials included neuromuscular measures (described below), $\dot{V}O_2$, respiratory exchange ratio

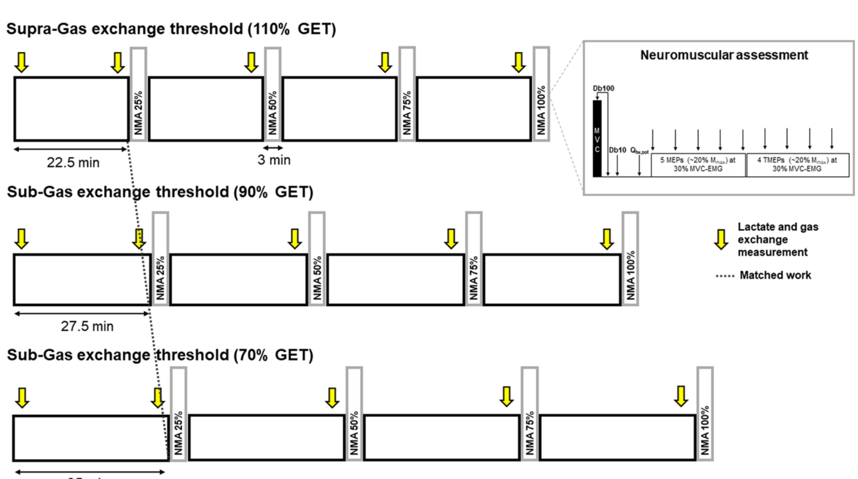

FIGURE 1—Schematic of constant work rate cycling exercise and neuromuscular assessment protocol. NMA, neuromuscular assessment; Db100, 100 Hz motor nerve stimulation; Db10, 10 Hz motor nerve simulation; $Q_{tw,pot}$, quadriceps potentiated twitch.

(RER), HR, and blood lactate. Three minutes after task completion, 2 min of baseline cycling was performed at 20 W, followed immediately by a ramp-incremental test with increments of 30 W·min^{-1} until task failure. Gas exchange and HR measurements were made throughout the ramp-incremental test, and blood lactate was measured 2 min after task failure. Outcome variables from the ramp-incremental tests performed after constant work rate exercise included $\dot{V}O_{2peak}$, peak power output, peak HR, and postexercise blood lactate. Two participants were unable to complete the postexercise ramp-incremental tests because of technical issues with the cycle ergometer, and the data were thus analyzed from 15 participants.

Neuromuscular function assessments. The neuromuscular assessments were performed on the right knee extensor and consisted of a maximal voluntary contraction (MVC) with a superimposed high-frequency doublet, high- and low-frequency doublets delivered by electrical stimulation to the motor nerve on relaxed muscles (Db100 and Db10, respectively), and a single potentiated twitch delivered by electrical stimulation on relaxed muscles ($Q_{tw,pot}$). To assess alterations occurring along the corticospinal–motoneuronal pathway, transcranial magnetic stimulation (TMS) was used to elicit motor-evoked potentials (MEP) for the assessment of corticospinal excitability, with thoracic electrical stimulation used to elicit thoracic MEP (TMEP) for the assessment of motoneuron excitability (Fig. 1). Both MEP and TMEP were evoked while participants contracted at the EMG level associated with 30% of the baseline MVC. The preexercise assessment began with the determination of the appropriate intensity for electrical nerve stimuli (described below). Afterward, a warm-up consisting of contractions at 25%, 50%, 75%, and 90% of perceived maximum force was performed. Subsequently, participants performed a ~3-s MVC without stimulation. At baseline, two MVC with a superimposed 100-Hz doublet were then performed, both followed by Db100, Db10, and $Q_{tw,pot}$ at 2, 4, and 6 s postcontraction, respectively. One minute separated all baseline MVC. Next, five single-pulse TMS were delivered to the hotspot at the EMG level associated with 30% MVC, followed by four thoracic electrical stimuli evoking TMEP at the same contraction intensity (described below). For neuromuscular assessments at 25%, 50%, 75%, and 100% of task completion, the same stimulus intensities were used, but only one MVC with superimposed and postcontraction stimuli was performed.

Experimental Procedures

Gas exchange measurements and capillary blood sample. From the breath-by-breath gas exchange and ventilation measurements, $\dot{V}O_2$ and RER were quantified. Before each visit, the MetaMax was calibrated, following manufacturer guidelines, to gases of known concentrations (O_2, 15.15%; CO_2, 5.03%). Ventilatory volumes were calibrated using a 3-L syringe. Moreover, both calibrations were reperformed before each measurement during the constant work rate trials. Capillary blood samples were taken from the fingertip and analyzed to measure lactate concentration (Lactate Scout, EKF Diagnostics, Cardiff, UK).

Recumbent cycle ergometer. Details on the innovative recumbent cycle ergometer have been described elsewhere (20). Briefly, the crank can be locked in place, with the foot secured and ankle position locked such that horizontal force production can be assessed. The seat position was adjusted to ensure that knee and hip angle were at 90° flexion when the pedal was locked. This position was recorded during the familiarization visit and used throughout the subsequent trials. While on the cycle ergometer, participants were secured at the chest with noncompliant straps. Isometric force was measured during voluntary and evoked contractions using a wireless PowerForce pedal force analysis system (Model PF1.0.0; Radlabor GmbH, Freiburg, Germany) (21) situated between the pedal and the crank. When the pedal was locked, the crank was parallel with the ground, with the ankle at 90° flexion. Force was sampled at 500 Hz and recorded using Imago Record (version 8.50, Radlabor GmbH). Visual force feedback was displayed on a large computer monitor in front of the participants, with the PowerForce signal transmitted to a PowerLab system (16/35; ADInstruments, Bella Vista, Australia) using a National Instruments 16-bit A/D card (NI PCI-6229; National Instruments, Austin, TX) and connector block (BNC-2111, National Instruments). The data recorded were subsequently analyzed offline using Labchart 8 software (ADInstruments).

Electromyography. Electromyographic activity was recorded during neuromuscular assessments and throughout cycling. Self-adhesive surface electrodes (10-mm recording diameter; Meditrace 100, Covidien, Mansfield, MA) were placed on the vastus lateralis (VL) of the right knee extensors using a bipolar configuration with a 30-mm interelectrode distance. A reference electrode was placed on the patella. Tape was placed on the electrode throughout cycling. Before applying the electrodes, the skin was shaved, gently abraded, and cleaned using isopropyl alcohol. The EMG signals were analog-to-digitally converted at a sampling rate of 2000 Hz using a PowerLab system (16/35, ADInstruments) and octal bioamplifier (ML138, ADInstruments, gain = 1000) with band-pass filter (10–500 Hz) and were analyzed offline using Labchart 8 software (ADInstruments). To determine the EMG level associated with 30% MVC, i.e., the intensity at which TMS and thoracic stimuli were delivered, participants held an isometric contraction at 30% MVC for 5 s during the baseline assessment for each constant work rate trial. The average smoothed rectified EMG was then calculated to derive a target EMG level for the task.

Femoral nerve stimulation. Single electrical stimuli (1-ms duration) were delivered to the right femoral nerve using a constant-current stimulator (DS7A; Digitimer Ltd., Welwyn Garden City, Hertfordshire, UK) with a 400-V maximal output voltage between surface electrodes. The cathode electrode (10 mm stimulating diameter; Meditrace 100, Covidien) was secured with tape on the inguinal triangle, and a 50 × 90-mm rectangular anode electrode (Durastick Plus, DJO Global, Vista, CA) was placed on the gluteal fold. The optimal stimulus intensity was determined as the minimum current that elicited a maximum resting twitch response (Q_{tw}) and compound muscle action potential (M_{max}). To do so, the initial intensity was set at 30 mA, with a 30 mA increment until a plateau in the contractile and EMG response. This intensity was subsequently multiplied by 1.3 to ensure the stimulus was supramaximal (122 ± 33, 126 ± 40, and 126 ± 29 mA for 110, 90, and 70% GET, respectively; repeated-measures ANOVA $P = 0.728$). This intensity was then used for the remainder of the constant work rate trial.

TMS. Single-pulse TMS of 1-ms duration was delivered to the left motor cortex to elicit MEP of the right VL with a concave double cone coil using a magnetic stimulator (Magstim 2002; The Magstim Co., Whitland, UK). The junction of the double cone coil was aligned tangentially to the sagittal plane, with its center 1–2 cm to the left of the vertex, and orientated to induce current in the posterior-to-anterior direction. During the familiarization visit, the hotspot was established as the position that elicited the largest MEP in the VL during a contraction at the EMG level associated with 30% MVC, with the intensity set at 40% of maximum stimulator output. Indelible marks were made at the hotspot and around the ears, and the cap was worn during subsequent trials to ensure a consistent stimulus location throughout the study. During all visits, the intensity required to elicit an MEP of 20%–30% M_{max} was determined (41% ± 9%, 43% ± 9%, and 42% ± 10% maximum stimulator output for 110%, 90%, and 70% GET, respectively; repeated-measures ANOVA $P = 0.199$). This intensity was maintained throughout the protocol for each of the respective trials.

Thoracic stimulation. Thoracic stimulations were delivered via a constant-current stimulator (DS7R, Digitimer Ltd.), evoking TMEP by passing a rectangular electrical pulse with a 1-ms duration and a 400-V maximal output voltage between surface electrodes. The anode was placed between the spinous processes of T_1 and T_2, and the cathode was positioned 5 cm below the center of the anode. Before the assessment, electrical stimuli were first administered at 50 mA and were then increased in 50 mA until a response of 20%–30% M_{max} was elicited (336 ± 148, 325 ± 154, and 300 ± 164 mA for 110%, 90%, and 70% GET, respectively; repeated-measures ANOVA $P = 0.06$). Two participants did not receive thoracic stimuli because of discomfort. Thus, results for TMEP and MEP/TMEP are reported from 15 participants.

Data Analysis

Gas exchange measurements. Gas exchange and ventilatory data were interpolated to 6-s intervals for subsequent analysis. $\dot{V}O_{2peak}$ was defined as the highest 30 s rolling average of $\dot{V}O_2$. The GET was determined using multiple gas exchange and ventilatory equivalent criteria through consideration of $\dot{V}O_2$, $\dot{V}CO_2$, end-tidal partial pressures for CO_2 and O_2, and $\dot{V}_E/\dot{V}CO_2$ and $\dot{V}_E/O_2$ (22). Each value was calculated independently by two experimenters and determined after they agreed on the value set. The power output associated with the GET was determined by linearly interpolating the $\dot{V}O_2$ versus power output relationship after the $\dot{V}O_2$ was left-shifted by a time interval corresponding to the MRT. For the MRT analysis, the average $\dot{V}O_2$ from the last 2 min of the 100-W step transition was calculated. Subsequently, a linear regression was used to fit the $\dot{V}O_2$ versus power output response, beginning from the onset of its systematic rise, as visually determined. The steady-state $\dot{V}O_2$ from the 100-W transition was then superimposed on the $\dot{V}O_2$ versus power output relationship. Next, the difference between the power output corresponding to the abscissa of the intersection between $\dot{V}O_2$ and the linear fit of $\dot{V}O_2$ versus power output and 100 W was determined and converted to time to yield the MRT (13.1 ± 5.2 W; 26.3 ± 10.4 s) (19). To assess whether a plateau in $\dot{V}O_2$ was evident at task failure during the ramps, the slope of the $\dot{V}O_2$–power output relationship was assessed from the beginning of the systematic rise in $\dot{V}O_2$, as visually determined, until 1 min before task failure. The $\dot{V}O_2$ was then predicted through extrapolation for the final 1 min. Participants met the criteria for a $\dot{V}O_2$ plateau if the actual $\dot{V}O_{2peak}$ was >100 $mL \cdot min^{-1}$ lower than the predicted $\dot{V}O_{2peak}$ (23). For the constant work rate trials, gas exchange and HR measurements were averaged from 3 to 4 min after exercise commencement and from 1 to 3 min before each neuromuscular assessment.

Force and electromyography. MVC was considered as the peak force during the 3-s MVC. To determine the maximum EMG, the root-mean-squared EMG (EMG_{RMS}) was calculated over a 0.2-s epoch around the peak torque and was subsequently normalized to M_{max} at the corresponding time point. The average VA, Db100, Db10:100, $Q_{tw,pot}$, and M_{max} from the two preexercise measurements were used as baseline values. For the assessment of VA, the following equation was used (24):

$$\%VA = 1 - \left(\frac{SIT}{Db100}\right)100$$

When stimuli were not delivered at the peak force, a correction was applied using the force at stimulation (F_{atstim}), the peak force, the size of the superimposed twitch (SIT), and the Db100, with the following equation applied (25):

$$\%VA_{corrected} = 1 - \left(SIT\frac{\left(\frac{F_{atstim}}{MVC}\right)}{(Db100)}\right)100$$

For evoked EMG response (M_{max}, MEP, and TMEP), the peak-to-peak amplitude was recorded. The average values of the five MEP and four TMEP at each time point were both normalized to M_{max} at the respective time point and used to assess corticospinal and spinal motoneuron excitability, respectively. To assess cortical excitability, the MEP/TMEP (%) was calculated at each time point. Prestimulus force and EMG_{RMS} were assessed using a 0.2-s epoch before the five TMS and the four thoracic stimuli, with the average of the nine values calculated at each time point. Prestimulus EMG was normalized to the maximum EMG measured during the preexercise MVC, and prestimulus force was normalized by the preexercise MVC force.

For measurements of EMG during cycling, the onset and the offset of the EMG burst in the VL during each pedal rotation were visually identified from the rectified signal, with EMG_{RMS} measured between these points. Cycling EMG was averaged at the same time points as gas exchange measurements, i.e., 3–4 min after commencing constant work rate cycling and 1–3 min before each neuromuscular assessment. The EMG_{RMS} was normalized to the preexercise maximum EMG_{RMS}.

Statistical Analysis

Jamovi statistical software (Jamovi, version 1.0, 2019, the jamovi project; retrieved from https://www.jamovi.org) was used for all statistical analyses. All data are presented as mean ± SD. Statistical significance was set at an α level of 0.05. Normality of the data was assessed by the Shapiro–Wilk test, with no data requiring transformation. Assumptions of sphericity were explored and controlled for all variables with the Greenhouse–Geisser adjustment, where necessary. For the comparisons between neuromuscular function, gas exchange variables, blood lactate, and HR during constant work rate exercise, a two-way repeated-measures ANOVA was used to assess the effect of exercise condition and time on dependent variables. For the ramp-incremental exercise tests, a one-way repeated-measures ANOVA was used, with selected variables ($\dot{V}O_{2peak}$, peak power output, peak HR, and postexercise lactate) compared between the incremental test performed during the initial visit (control ramp) and after the constant work rate trials. In the event of a significant main effect or interaction (condition–time), a Tukey *post hoc* analysis was performed to locate where differences lay. Partial eta squared (η_p^2) was calculated to estimate effect sizes, with values representing small ($\eta_p^2 < 0.13$), medium ($\eta_p^2 \geq 0.13$, <0.26), and large ($\eta_p^2 \geq 0.26$) effects.

RESULTS

The $\dot{V}O_{2peak}$ from the initial ramp-incremental test was 56.2 ± 10.5 $mL \cdot kg \cdot min^{-1}$ (3.9 ± 0.5 $L \cdot min^{-1}$), with peak power output 333 ± 49 W. The GET occurred at a $\dot{V}O_2$ of 31.7 ± 5.9 $mL \cdot kg^{-1} \cdot min^{-1}$ (53.8% ± 5.9% $\dot{V}O_{2peak}$), corresponding with a power output of 149 ± 35 W (44.5% ± 6.4% peak power output). Thus, the power outputs for the 110%, 90%, and 70% GET trials were 164 ± 39 W, 134 ± 32 W, and 105 ± 25 W, corresponding with a target work of 881 ± 216 kJ. The $\dot{V}O_2$ and the blood lactate data indicate that the target intensities were well met, with a significant increase in $\dot{V}O_2$ at 25% task completion confirming the presence of a $\dot{V}O_{2SC}$ only for the 110% GET trial and blood lactate increasing only for the 110% GET trial (Fig. 2). At 3–4 min of the 110% GET trial, $\dot{V}O_2$ was 111% ± 11% of the GET, with 15 out of 17 exercising above the GET. For the two participants whose $\dot{V}O_2$ was slightly below the GET during the 110% GET trial, an increase in blood lactate of >0.5 mM and in $\dot{V}O_2$ >50 $mL \cdot min^{-1}$ was observed at 25% of task completion, indicating that they were exercising within the heavy-intensity domain. For the 90% GET trial, $\dot{V}O_2$ was 93% ± 11% of the GET at 3–4 min, with 14 out of 17 participants exercising below the GET. For the three participants whose $\dot{V}O_2$ was slightly above the GET, only one showed an increase in blood lactate >0.5 mM, but the same participant displayed no clear slow component (increase in $\dot{V}O_2$ <10 $mL \cdot min^{-1}$ at 25% task

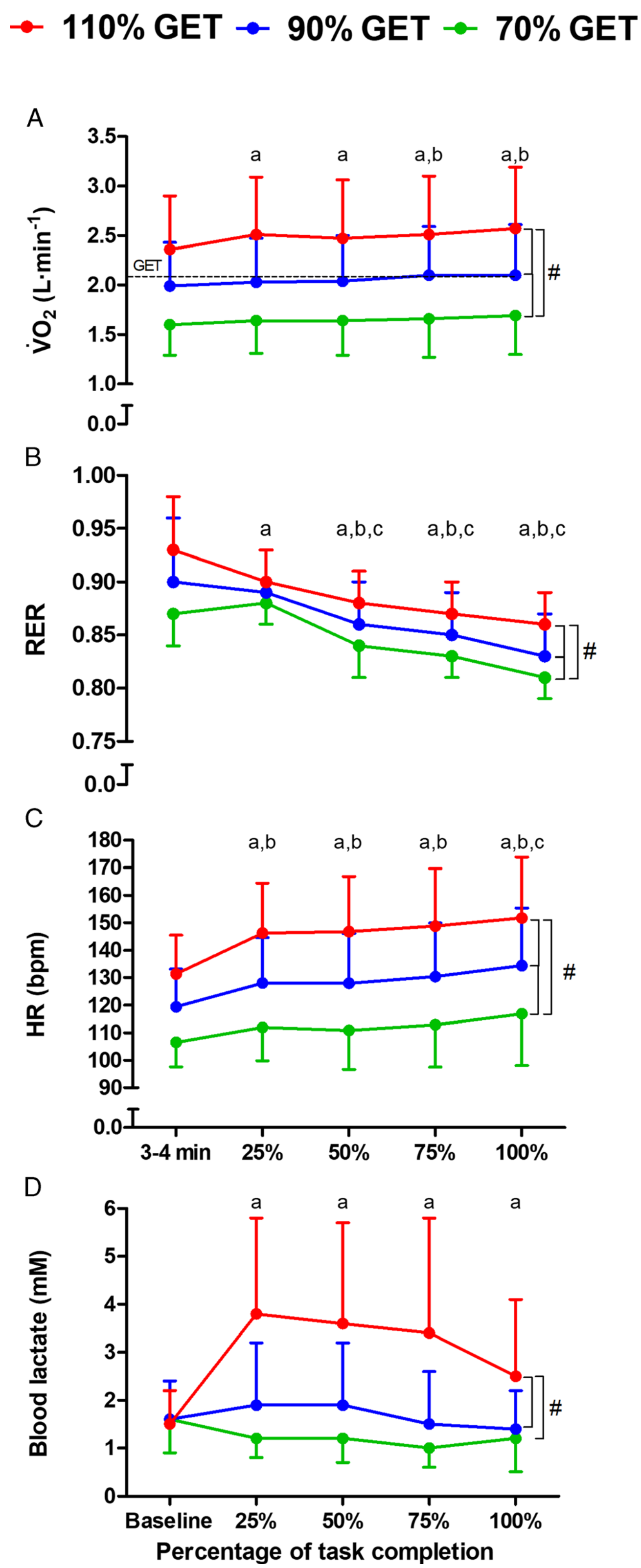

FIGURE 2—Pulmonary oxygen uptake ($\dot{V}O_2$, A), RER (B), HR (C), and blood lactate (D) measured during matched-work cycling at 110%, 90%, and 70% of GET. The dashed line in panel A displays the GET. a, significantly different from 3 to 4 min (for panels A, B, and D) or baseline (for panel C) for 110% GET; b, significantly different from 3 to 4 min for 90% GET; c, significantly different from 3 to 4 min for 70% GET. # Main effect of condition (P < 0.05).

completion). One participant reached task failure with 7 min remaining during the 110% GET trial. In this participant, the 110% GET trial had been performed first. Thus, the duration of the 90% and 70% trials was reduced by 8.5 and 11 min, respectively, to ensure that the trials remained work matched.

Gas exchange, HR, and blood lactate during constant work rate trials. An increase in $\dot{V}O_2$ between 3–4 min and 1–3 min before the 25% neuromuscular assessment ($P < 0.001$) confirmed the presence of a $\dot{V}O_{2SC}$ for the 110% GET trial only (time–condition interaction: $F_{3.7,58.9} = 3.4$, $P = 0.016$, $\eta_p^2 = 0.918$). For the 90% GET trial, $\dot{V}O_2$ remained stable until 75% and 100% of task completion, when it increased relative to 3–4 min (both $P \leq 0.025$). No change in $\dot{V}O_2$ was found for the 70% GET trial (all $P \geq 0.206$; Fig. 2A). The decrease in RER occurred more rapidly for the 110% GET (time–condition interaction: $F_{4.6,72.8} = 2.2$, $P = 0.029$, $\eta_p^2 = 0.12$), being reduced from 25% to 100% of task completion (all $P \leq 0.017$), whereas RER was decreased from 50% to 100% of task completion for the 90% and 70% GET trials (all $P < 0.001$; Fig. 2B). The change in HR differed between conditions (time–condition interaction: $F_{3.4,54.4} = 10.0$, $P < 0.001$, $\eta_p^2 = 0.38$), with *post hoc* tests showing that HR increased relative to 3–4 min at all time points when cycling at 110% and 90% GET (all $P \leq 0.021$), whereas HR increased at 100% task completion at 70% GET (all $P < 0.001$; Fig. 2C). Blood lactate increased only during the 110% GET trial (time–condition interaction: $F_{4.3,69.5} = 7.7$, $P < 0.001$, $\eta_p^2 = 0.32$), with *post hoc* tests showing that blood lactate was increased relative to 20 W baseline cycling at all time points for the 110% GET trial (all $P < 0.001$), but blood lactate remained unchanged throughout the 90% and 70% GET trials (all $P \geq 0.779$; Fig. 2D).

Voluntary and evoked force. Changes in voluntary and evoked force are displayed in Figure 2, with individual data displayed in Supplemental Figure 1 (see Supplemental Digital Content, http://links.lww.com/MSS/C654). Although voluntary and evoked forces decreased throughout all constant work rate trials (time effects all $P \leq 0.014$; Fig. 3), the magnitude of the

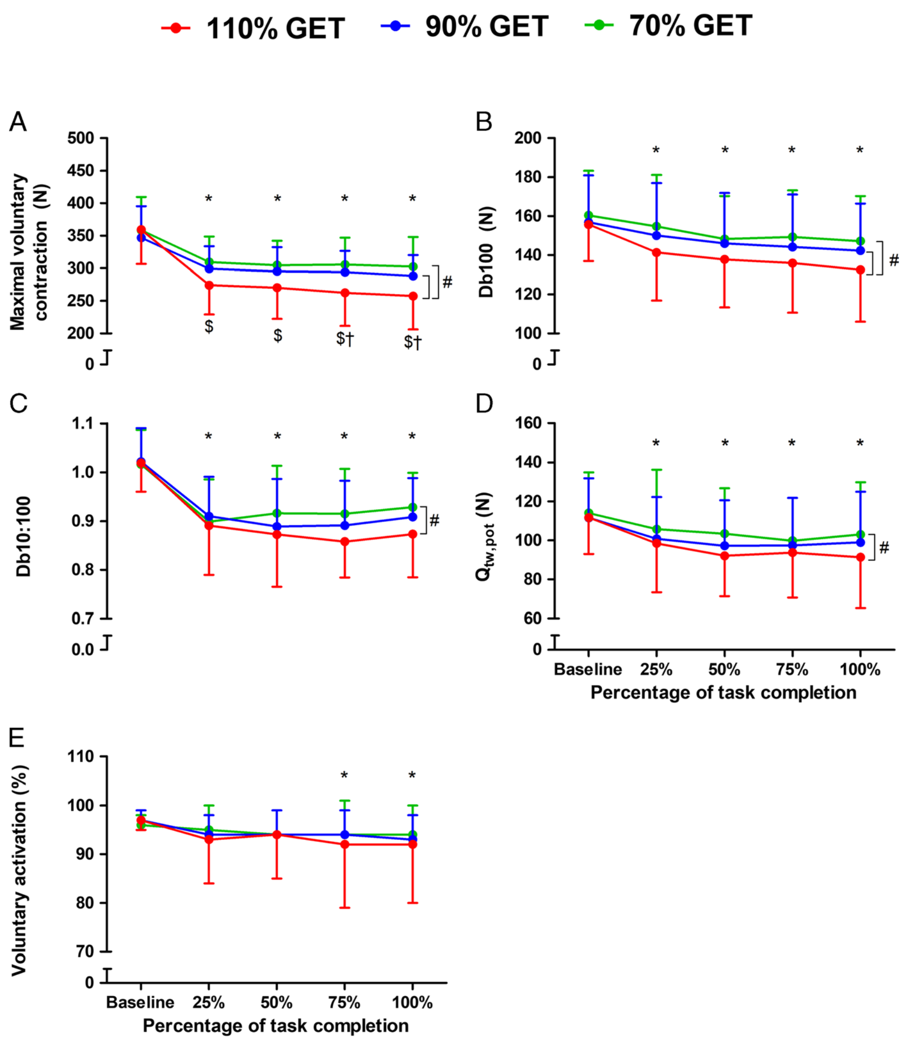

FIGURE 3—MVC (A), high-frequency doublet (Db100, B), low- to high-frequency doublet force ratio (Db10:100, C), potentiated twitch force ($Q_{tw,pot}$, D), and VA (E) during matched-work cycling at 110%, 90%, and 70% of GET. #Main effect of condition ($P < 0.05$). *Significantly different from baseline ($P < 0.05$). $Significantly lower for 110% vs 70% GET ($P < 0.05$). †Significantly lower for 110% vs 90% GET ($P < 0.05$). Individual data are presented in Supplemental Figure 1 (see Supplemental Digital Content, http://links.lww.com/MSS/C654).

APPLIED SCIENCES

decrease in MVC (condition effect: $F_{2,32}$ = 13.08, P < 0.001, η_p^2 = 0.45) and Db100 (condition effect: $F_{2,32}$ = 5.79, P < 0.001, η_p^2 = 0.27) was dependent on the condition. Specifically, MVC was lower during 110% than 90% (P = 0.008) and 70% GET (P < 0.001), but no difference was found between 90% and 70% GET (P = 0.175; Fig. 3A). Similarly, Db100 was lower during 110% than 90% (P = 0.029) and 70% GET (P = 0.003), with no difference between 90% and 70% GET (P = 0.431; Fig. 3B). The Db10:100 also depended on the condition (condition effect: $F_{2,32}$ = 3.46, P = 0.044, η_p^2 = 0.19) and was lower throughout the 110% GET trial relative to 70% GET (P = 0.037), with no difference between 110% and 90% GET (P = 0.217) or between 90% and 70% GET (P = 0.659; Fig. 3C). Similarly, $Q_{tw,pot}$ differed between conditions (condition effect: $F_{2,32}$ = 5.51, P < 0.014, η_p^2 = 0.26), with $Q_{tw,pot}$ lower throughout the 110% versus 70% GET trial (P = 0.006) and no difference between 110% and 90% (P = 0.257) or between 90% and 70% GET trials (P = 0.217; Fig. 3D). The reduction in VA did not differ between the three trials (condition effect: $F_{1.3,20.0}$ = 0.39, P = 0.580, η_p^2 = 0.02). Throughout all trials, the reductions in MVC and resting evoked twitch responses were observed at 25% of task completion, with no further decrease occurring thereafter (P > 0.05). By contrast, VA was reduced only at 75% and 100% of task completion ($P \leq 0.008$). Time–condition interactions are displayed where appropriate in Figure 3.

Voluntary and evoked electromyography. During constant work rate exercise, M_{max}, MEP, and TMEP were reduced (time effects all P < 0.001), whereas maximum EMG_{RMS} and MEP/TMEP were unchanged (time effect $P \geq 0.052$; Fig. 4). All reductions in M_{max}, MEP, and TMEP amplitude were observed at 25% of task completion, with no further decrease occurring thereafter (all P > 0.05). The changes in these variables were similar between conditions, with no main effects of condition or time–condition interactions (all P > 0.05). Prestimulus EMG remained stable throughout the constant work rate trials (time effect P = 0.871), whereas prestimulus force was reduced at all time points (all P < 0.001). These variables did not differ between conditions (P > 0.05). The EMG_{RMS} during cycling did not change over time for any condition (P > 0.05). However, EMG_{RMS} was

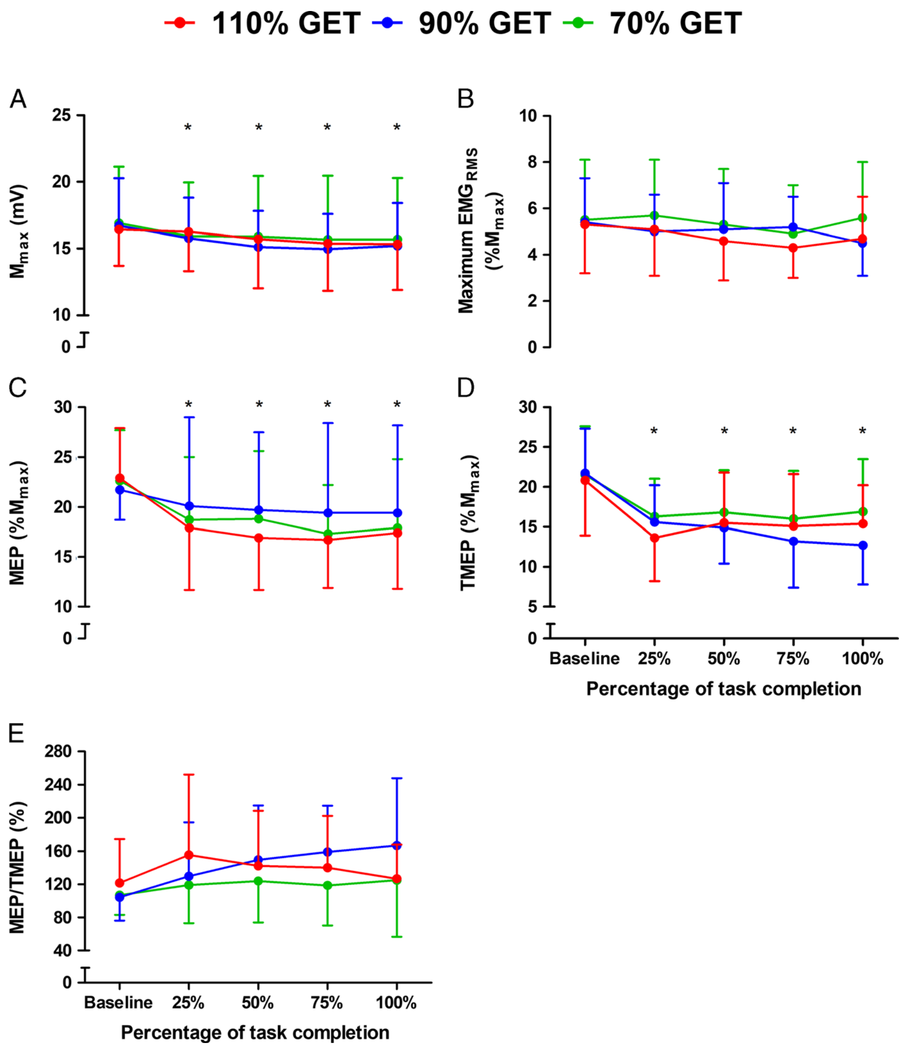

FIGURE 4—Maximal compound muscle action potential (M_{max}, A), maximum EMG_{RMS} (B), MEP amplitude (C), TMEP amplitude (D), and the MEP/TMEP percentage ratio (E) measured during matched-work cycling at 110%, 90%, and 70% of GET. *Significantly different from baseline (P < 0.05).

higher for 110% compared with 90% and 70% GET (all P < 0.001) and higher for 90% compared with 70% GET (P = 0.014; Fig. 5).

Postexercise ramp-incremental tests. Variables from the control ramp performed in an unfatigued state and post–ramp-incremental tests are displayed in Figure 6. Criterion for a $\dot{V}O_2$ plateau was met in 53%, 67%, 63%, and 63% of participant for the control, 110%, 90%, and 70% GET ramps, respectively. The $\dot{V}O_{2peak}$ was lower after the 110% GET trial relative to the control ramp, being reduced by 266 ± 355 mL·min^{-1} (i.e., 6.5% ± 9.3%; P = 0.008). No differences in $\dot{V}O_{2peak}$ were found between the control ramp and the ramp tests performed at 90% (P = 0.267) or 70% GET (P = 0.889). Furthermore, $\dot{V}O_{2peak}$ was 211 ± 347 mL·min^{-1} lower (i.e., 5.0% ± 8.9%) for the 110% compared with the 70% trial (P = 0.048), with no difference between 110% and 90% (P = 0.410) or 90% and 70% GET (P = 0.675; Fig. 6A). Similarly, peak power output was 24 ± 39 W (i.e., 7.0% ± 11.3%) lower after the 110% GET trial relative to the control ramp (P = 0.015), with no other between-trial differences (P ≥ 0.078; Fig. 6B). Peak HR was not different between conditions ($F_{3,42}$ = 0.66, P = 0.582, η_p^2 = 0.05; Fig. 6C). Blood lactate was lower after the three constant work rate trials relative to the control ramp performed without prior exercise (all P ≤ 0.032), with no other between-trial differences (P ≥ 0.089; Fig. 6C).

DISCUSSION

The primary aim of the present study was to assess performance fatigability during work-matched exercise marginally above and below the GET. When exercising at 70% or 90% GET, no differences were observed in fatigability for a given level of work, with decreases in MVC and evoked resting twitch responses comparable between the two work rates. However, when the power output was elevated by the same magnitude between 90% and 110% GET, thus transcending the moderate-heavy domain boundary, fatigability was substantially exacerbated. Thus, completing a given amount of work within the moderate domain results in the same fatigability regardless of work rate, whereas completing the same amount of work within the lower boundaries of the heavy domain results in considerably greater fatigability. This indicates that even marginally exceeding the moderate-heavy domain boundary has negative implications for neuromuscular function, with evoked twitch responses to relaxed muscle indicating that the greater fatigability is due to intramuscular mechanisms. A secondary aim of this study was to assess performance during ramp-incremental exercise performed after sub- and supra-GET constant work rate exercise. The results displayed that $\dot{V}O_{2peak}$ and peak power output were only impaired with respect to the control ramp when performed after the 110% GET trial, with no changes after the 90% or 70% GET trials. Together, these results meet our hypotheses and suggest that the GET represents a boundary above which fatigability and impairments in exercise tolerance are exacerbated.

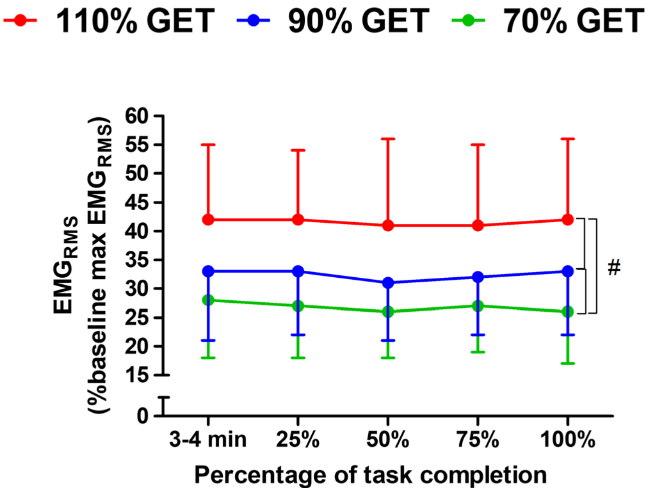

FIGURE 5—EMG$_{RMS}$ measured during matched-work cycling at 110%, 90%, and 70% of GET, normalized to baseline maximum EMG$_{RMS}$. #Main effect of condition (P < 0.05).

Intramuscular mechanisms induce greater fatigability above the GET. The present study showed that fatigability was substantially greater for a given level of work when exercising in the heavy domain (i.e., above the GET) relative to the moderate domain (i.e., below the GET). At task completion, the reduction in MVC was −27% ± 17% for 110% GET, −16% ± 7% for 90% GET, and −15% ± 10% for 70% GET. Thus, the reduction in MVC was up to 1.8-fold greater for a given level of work when exercising above relative to below the GET. Several results indicate that factors residing within the muscle were responsible for the greater fatigability during supra-GET exercise. First, the reduction in the potentiated doublet was greater at 110% GET relative to the 90% and 70% trials, being reduced by −16% ± 10%, −9% ± 9%, and −7% ± 11% at task completion, respectively. Second, the temporal pattern of change between the MVC and the potentiated doublet was similar. Third, although reductions in VA at 75% and 100% of task completion were significant, they were similarly modest (≤5%) in all conditions. Finally, although the reductions in M_{max} indicate a possible contribution of impaired sarcolemma excitability to attenuated force production, M_{max} was reduced similarly between intensities. Overall, our findings suggest that transcending the moderate- to heavy-intensity domain boundary has negative implications for contractile function.

Greater fatigability manifests early during heavy-intensity exercise. The observation of the temporal pattern of change in voluntary and evoked force responses provides mechanistic insight into the greater reduction in contractile function above the GET in the present study. Specifically, the majority of the force loss throughout all trials occurred within the first 25% of the task. For example, at 25% of task completion, MVC was reduced by −23% ± 14%, −12% ± 7%, and −13% ± 7% for the 110%, 90%, and 70% GET trials, respectively, with no significant force loss occurring thereafter. Similarly, the reduction in Db100 at 25% task completion was −11% ± 9%, −4% ± 10%, and −4% ± 9% for the 110%, 90%, and 70% GET trials, respectively, again with no significant additional force loss from 25% to 100% of task completion. Thus, it appears that intramuscular processes occurring

APPLIED SCIENCES

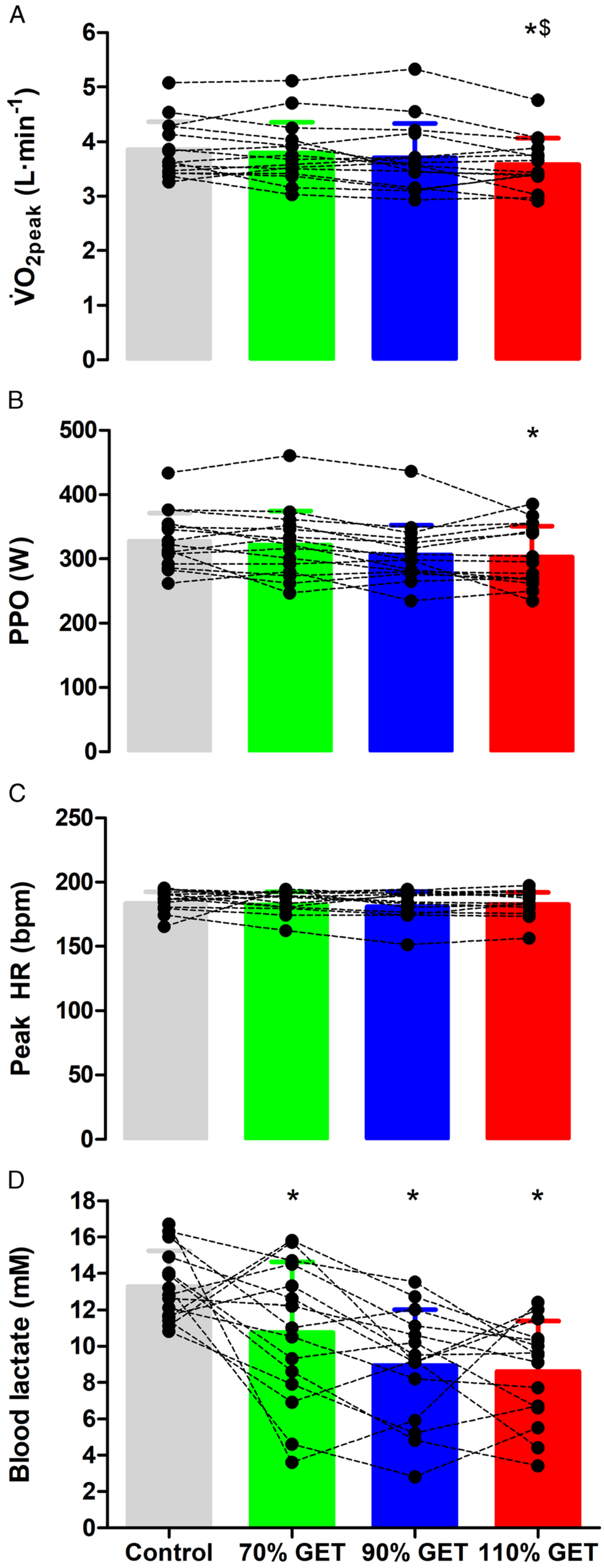

FIGURE 6—Pulmonary oxygen uptake ($\dot{V}O_2$, A), peak power output (PPO, B), peak HR (C), and blood lactate (D) measured in response to ramp-incremental exercise performed in an unfatigued state (baseline) or after matched-work exercise at 110%, 90%, and 70% GET. *Black symbols* and *dashed lines* show individual data. *Significantly different from baseline (*P* < 0.05). $Significantly different from 70% GET trial (*P* < 0.05).

early during exercise were largely responsible for the greater fatigability when exercising above versus below the GET.

An important difference in the physiological response to supra- versus sub-GET exercise is the emergence of a $\dot{V}O_{2SC}$ in the early stages of exercise when cycling above the GET. The presence of a $\dot{V}O_{2SC}$ was confirmed in the present study by the increase in $\dot{V}O_2$ between 3 and 4 min of exercise and 25% of task completion (i.e., 22.5 min) during the 110% GET trial (by 151 ± 133 mL·min^{-1}) with no change at the same time point for the 90% and 70% GET trials. Although the mechanisms of the $\dot{V}O_{2SC}$ are incompletely understood (26), fatigability is thought to be essential in its development (12,16,17). For example, de Almeida Azevedo et al. (17) recently demonstrated that impairments in contractile function are associated with the amplitude of the $\dot{V}O_{2SC}$ across various heavy and severe exercise intensities and durations, with similar results displayed by Keir et al. (16) in response to severe-intensity cycling. Two prevailing hypotheses concerning the link between fatigability and $\dot{V}O_{2SC}$ are 1) impairments in contractile function of early recruited motor units necessitates the recruitment of additional, higher threshold motor units, thereby increasing O_2 demand (27–29), and 2) exceeding a critical limit of metabolic instability during the initial stages of exercise impairs contractile function and efficiency in already recruited fibers through common mechanisms (26,30). The former hypothesis is not supported in the present study given the stability of VL EMG_{RMS} in the presence of a manifested $\dot{V}O_{2SC}$ during supra-GET cycling. Although the limitations of compound EMG might limit the ability to detect subtle increases in motor unit recruitment (31), the lack of temporal association between the EMG_{RMS} and the $\dot{V}O_{2SC}$ corroborates numerous other studies (32–34) but conflicts with others (27,28). Regarding the latter hypothesis, it is interesting to note that in the present study, both heavy- and moderate-intensity exercise elicited impairments in neuromuscular function in contrast to previous results (12). However, it is possible that during heavy-intensity cycling, "critical limits" of metabolic instability and fatigability were exceeded, thereby impairing efficiency, driving the increase in ATP and O_2 demand, and further increasing the concentration of metabolites which impair contractile function (35–37). Although speculative, in light of the evidence that fatigability is implicit in inducing the $\dot{V}O_{2SC}$ (12), the greater level of fatigability during the 110% GET trial might have been evident during the initial minutes of cycling and subsequently led to the development of the $\dot{V}O_{2SC}$.

The role of glycogen depletion in impairing excitation–contraction coupling function has been established (38). Given the elevated glycolysis, as indicated by the higher RER, and the disproportionately higher ATP demand when exercising in the heavy versus moderate domain (18), the GET likely represents an important threshold above which the rate of glycogen depletion is exacerbated (2). Although glycogen stores would have fallen throughout the constant work rate trials in the present study, as indicated by the drop in RER and the decrease in postramp blood lactate (discussed below), the kinetics of reduced neuromuscular function in the present study are not indicative of a substantial contribution of glycogen to fatigability. For example, although low-frequency force depression was evident at all exercise intensities, the drop in the Db10:100 ratio occurred at 25% of task completion and did not further decrease thereafter. Given that sarcoplasmic reticulum Ca^{2+} release is impaired when intramyofibrillar glycogen stores reach critically low levels (39), a further drop in Db10:100 would be expected in the latter stages of exercise where glycogen depletion would have been implicated. Thus, it is possible that the duration of exercise in the present study was not long enough for glycogen to fall below critical levels in most participants. However, it should be noted that, to our knowledge, no study assessing the kinetics of altered neuromuscular function using neurostimulation methods *in vivo* has demonstrated a temporal pattern of impaired contractile function consistent with a contribution of glycogen depletion (40,41). Thus, the observed low-frequency force depression was more likely due to the known effects of metabolite accumulation (particularly Pi and H^+) on Ca^{2+} release/sensitivity rather than glycogen depletion (42,43).

Moderate-intensity exercise elicits impairments in neuromuscular function. A further interesting and novel finding from the present study was that, although fatigability was lower in the moderate than heavy domain, impairments in neuromuscular function were nevertheless evident during moderate-intensity exercise after just 25% of task completion. These results are in contrast to those of Cannon et al. (12), who found no change in isokinetic peak power after 8 min of moderate-intensity exercise. This might relate to 1) differences in the contraction modality used to assess fatigability (44) (i.e., dynamic in Cannon et al. [12] vs isometric in the present study) and/or 2) differences in exercise duration (8 min in Cannon et al. [12] vs 25% of task completion in the present study, which was 27.5 and 35 min for the 90% and 90% GET trials, respectively). Although metabolite accumulation does not occur progressively during moderate-intensity exercise, the concentration of metabolites such as Pi is nevertheless elevated with respect to rest (14,45). In turn, metabolites such as Pi have potent effects on the excitation–contraction coupling and/or cross-bridge force (46). Thus, even brief bouts of moderate-intensity exercise are associated with some degree of neuromuscular fatigability.

Spinal mechanisms contribute to impaired central nervous system function during moderate- and heavy-intensity exercise. In the present study, reductions in VA of 3%–5% were observed at 75% and 100% of task completion for all constant work rate trials. The reductions in VA toward the latter stages of exercise corroborate numerous other studies showing delayed impairments in VA during prolonged locomotor exercise (41,47,48). To provide segmented insight into perturbations occurring along the corticospinal–motoneuronal pathway, the present study assessed MEP and TMEP at each time point using motor cortical and thoracic spinal stimulations, respectively. Both MEP and TMEP were reduced at each time point during all constant work rate trials, with their similar magnitude of reduction indicating that impairments in corticospinal excitability were due to a reduced capacity of alpha-motoneurons to respond to synaptic input. The 18%–35%

reduction in TMEP in the present study was slightly greater than the nonsignificant 15% reduction found after 3 h of cycling in our recent study (49). This is likely due to the delay in the neuromuscular assessment in the latter study, given that previous work has shown that motoneuron excitability recovers rapidly after isometric tasks (50). A plausible mechanism of reduced motoneuron excitability relates to intrinsic motoneuron adaptations in response to repetitive activation. For example, previous studies in response to isometric exercise of both the upper and the lower limbs indicate that reductions in motoneuron excitability occur early during exercise and are most prominent in the most active motoneurons (50–52), possibly owing to late motoneuron spike frequency adaptation and/or reductions in persistent inward currents. Although the implications of altered, evoked EMG responses are difficult to delineate, reductions in motoneuron excitability have potentially important consequences for exercise tolerance given that they would likely necessitate a greater descending input to maintain motoneuron output and muscle activation (53). Thus, these results provide important, novel insight into corticospinal–motoneuronal perturbations during prolonged endurance exercise, with potential implications for exercise tolerance.

Maximal exercise capacity is impaired after exercise above the GET. To assess the effect of work-matched moderate- and heavy-intensity exercise on subsequent performance, the present study compared performance from postexercise ramp tests between the constant work rate trials and the baseline ramp test performed in a nonfatigued state. The results displayed that performance was only impaired relative to the baseline test after heavy-intensity and not moderate-intensity exercise. Specifically, peak power output was reduced by 7.0% ± 11.3% after the 110% GET trial, with no change after the 90% or 70% GET trials. Based on our data, there are at least three potentially important contributors to this finding. First, although we would expect a large component of metabolite accumulation to have subsided during the recovery period after constant work rate exercise and the low-intensity cycling at the beginning of the ramp (54), the ramp exercise might have nevertheless been initiated with higher intramuscular metabolite concentrations after the 110% GET trial relative to the moderate-intensity trials. In turn, this could have led to the quicker attainment of the consistent levels of metabolic disturbance, which are associated with task failure during severe-intensity exercise (11,55). Second, an interesting finding from the present study was that, concurrent with the decrease in peak power output, $\dot{V}O_{2peak}$ was reduced by 6.5% ± 9.3% after the 110% GET trial, with this $\dot{V}O_{2peak}$ lower than that associated with both the baseline and the 70% GET trial. Although the mechanism(s) contributing to the reduction in $\dot{V}O_{2peak}$ after heavy-intensity exercise is unclear, any impairment in the capacity for oxidative phosphorylation would necessitate an increase in substrate level phosphorylation to meet ATP demands and, thus, a potentially more rapid accumulation of metabolites deleterious to contractile function. Third, the reduced postramp blood lactate after the constant work rate trials suggests that the capacity for ATP production via anaerobic glycolysis was diminished, likely due to glycogen depletion. In turn, glycogen depletion is associated with a reduction in the tolerable degree of exercise within the severe-intensity domain (W′) (56,57) and, thus, might have contributed to the reduction in peak power output. Together, the impairment in the capacity for oxidative and nonoxidative ATP resynthesis combined with the greater impairments in neuromuscular function might have been responsible for impaired exercise performance after heavy-intensity exercise.

Regarding the reduction in $\dot{V}O_{2peak}$ after the heavy-intensity trial, it is uncertain whether this reflects a genuine reduction in $\dot{V}O_{2max}$ or was simply the result of lower motivation and an earlier task disengagement. Although some use a "supramaximal" verification trial after ramp-incremental exercise to confirm (or refute) the attainment of $\dot{V}O_{2max}$ (14), such an approach would have been unsuitable in the present study given that a recovery period of 5–20 min is regularly used (14,58) between task failure after the ramp and commencing the validation trial. Moreover, recent work has highlighted methodological limitations associated with supramaximal verification trials (23). The finding that $\dot{V}O_{2peak}$ was the same for the control ramp and the ramps after the 70% and 90% GET trials provides support for $\dot{V}O_{2max}$ being attained during these trials. Although we cannot rule out an early disengagement during the ramp after the 110% GET trial, peak HR from the incremental tests was consistent across all trials, whereas the percentage of participants meeting the criteria for a plateau in $\dot{V}O_2$ during the 110% GET trial ramp (67%) was higher than the other ramps (53%–63%). Although these results appear favorable in indicating that $\dot{V}O_{2max}$ was attained, there remains some degree of uncertainty, and future studies could determine whether this result is repeatable.

CONCLUSIONS

By performing regular assessments of neuromuscular function during work-matched exercise at three equidistant work rates above (110%) and below (70% and 90%) the GET, the present study shows that fatigability was exacerbated for a given amount of work when exercising above the GET, whereas no differences were observed at work rates below the GET. These results suggest that transcending the boundary between the moderate- and the heavy-intensity domain has negative implications for neuromuscular function. The greater reduction in resting evoked twitch responses when exercising above the GET, concurrent with the lack of difference between intensities for voluntary activation (VA) and sarcolemma excitability, suggests that the greater fatigability is due to mechanisms residing within the muscle. Finally, maximal exercise capacity measured after constant work rate moderate- and heavy-intensity exercise was only impaired after exercise above the GET. This might relate to the greater neuromuscular impairment accrued throughout the heavy-intensity trial, as well as the impairment in peak oxygen uptake and capacity for anaerobic glycolysis. These results further emphasize the importance of exercise intensity domains in determining the neuromuscular response to exercise.

The authors thank the participants for their commitment during the study. No funding was received for the present study.

The results of the present study do not constitute endorsement by the American College of Sports Medicine. The results of the study are presented clearly, honestly, and without fabrication, falsification, or inappropriate data manipulation.

No conflicts of interest, financial or otherwise, are declared by the authors.

Data are available upon request.

C. G. B. conceived the study; C. G. B., F. S. P., and G. Y. M. designed the study; C. G. B., F. S. P., and J.M. performed the experiments; C. G. B. analyzed the data; C. G. B. interpreted the results of experiment; C. G. B. drafted manuscript; C. G. B., F. S. P., J. M., J. M. M., and G. Y. M. edited and revised the manuscript; C. G. B., F. S. P., J. M., J. M. M., and G. Y. M. approved the final version of manuscript.

REFERENCES

1. Hawley JA, Hargreaves M, Joyner MJ, Zierath JR. Integrative biology of exercise. *Cell*. 2014;159(4):738–49.
2. Burnley M, Jones AM. Oxygen uptake kinetics as a determinant of sports performance. *Eur J Sport Sci*. 2007;7(2):63–79.
3. Poole DC, Burnley M, Vanhatalo A, Rossiter HB, Jones AM. Critical power: an important fatigue threshold in exercise physiology. *Med Sci Sports Exerc*. 2016;48(11):2320–34.
4. Iannetta D, Ingram CP, Keir DA, Murias JM. Methodological reconciliation of CP and MLSS and their agreement with the maximal metabolic steady state. *Med Sci Sports Exerc*. 2022;54(4):622–32.
5. Enoka RM, Duchateau J. Translating fatigue to human performance. *Med Sci Sports Exerc*. 2016;48(11):2228–38.
6. Millet GY, Martin V, Martin A, Vergès S. Electrical stimulation for testing neuromuscular function: from sport to pathology. *Eur J Appl Physiol*. 2011;111(10):2489–500.
7. Jones AM, Wilkerson DP, DiMenna F, Fulford J, Poole DC. Muscle metabolic responses to exercise above and below the "critical power" assessed using 31P-MRS. *Am J Physiol Regul Integr Comp Physiol*. 2008;294(2):R585–93.
8. Sundberg CW, Fitts RH. Bioenergetic basis of skeletal muscle fatigue. *Curr Opin Physiol*. 2019;10:118–27.
9. Burnley M, Vanhatalo A, Jones AM. Distinct profiles of neuromuscular fatigue during muscle contractions below and above the critical torque in humans. *J Appl Physiol (1985)*. 2012;113(2):215–23.
10. Azevedo RA, Forot J, Iannetta D, MacInnis MJ, Millet GY, Murias JM. Slight power output manipulations around the maximal lactate steady state have a similar impact on fatigue in females and males. *J Appl Physiol (1985)*. 2021;130(6):1879–92.
11. Black MI, Jones AM, Blackwell JR, et al. Muscle metabolic and neuromuscular determinants of fatigue during cycling in different exercise intensity domains. *J Appl Physiol (1985)*. 2017;122(3):446–59.
12. Cannon DT, White AC, Andriano MF, Kolkhorst FW, Rossiter HB. Skeletal muscle fatigue precedes the slow component of oxygen uptake kinetics during exercise in humans. *J Physiol*. 2011;589(Pt 3):727–39.
13. Brownstein CG, Twomey R, Temesi J, Medysky ME, Culos-Reed SN, Millet GY. Mechanisms of neuromuscular fatigability in people with cancer-related fatigue. *Med Sci Sports Exerc*. 2022;54(8):1355–63.
14. Korzeniewski B. P(i)-induced muscle fatigue leads to near-hyperbolic power-duration dependence. *Eur J Appl Physiol*. 2019;119(10):2201–13.
15. Rossiter HB, Ward SA, Kowalchuk JM, Howe FA, Griffiths JR, Whipp BJ. Dynamic asymmetry of phosphocreatine concentration and O(2) uptake between the on- and off-transients of moderate- and high-intensity exercise in humans. *J Physiol*. 2002;541(Pt 3):991–1002.
16. Keir DA, Copithorne DB, Hodgson MD, Pogliaghi S, Rice CL, Kowalchuk JM. The slow component of pulmonary O_2 uptake accompanies peripheral muscle fatigue during high-intensity exercise. *J Appl Physiol (1985)*. 2016;121(2):493–502.
17. de Almeida Azevedo R, Keir DA, Forot J, Iannetta D, Millet GY, Murias JM. The effects of exercise intensity and duration on the relationship between the slow component of $\dot{V}O_2$ and peripheral fatigue. *Acta Physiol (Oxf)*. 2022;234(2):e13776.
18. Cannon DT, Bimson WE, Hampson SA, et al. Skeletal muscle ATP turnover by 31P magnetic resonance spectroscopy during moderate and heavy bilateral knee extension. *J Physiol*. 2014;592(23):5287–300.
19. Iannetta D, Murias JM, Keir DA. A simple method to quantify the $\dot{V}O_2$ mean response time of ramp-incremental exercise. *Med Sci Sports Exerc*. 2019;51(5):1080–6.
20. Doyle-Baker D, Temesi J, Medysky ME, Holash RJ, Millet GY. An innovative ergometer to measure neuromuscular fatigue immediately after cycling. *Med Sci Sports Exerc*. 2018;50(2):375–87.
21. Stapelfeldt B, Mornieux G, Oberheim R, Belli A, Gollhofer A. Development and evaluation of a new bicycle instrument for measurements of pedal forces and power output in cycling. *Int J Sports Med*. 2007;28(4):326–32.
22. Beaver WL, Wasserman K, Whipp BJ. A new method for detecting anaerobic threshold by gas exchange. *J Appl Physiol (1985)*. 1986;60(6):2020–7.
23. Iannetta D, de Almeida Azevedo R, Ingram CP, Keir DA, Murias JM. Evaluating the suitability of supra-PO(peak) verification trials after ramp-incremental exercise to confirm the attainment of maximum O(2) uptake. *Am J Physiol Regul Integr Comp Physiol*. 2020;319(3):R315–22.
24. Merton PA. Voluntary strength and fatigue. *J Physiol*. 1954;123(3):553–64.
25. Strojnik V, Komi PV. Neuromuscular fatigue after maximal stretch-shortening cycle exercise. *J Appl Physiol (1985)*. 1998;84(1):344–50.
26. Korzeniewski B, Zoladz JA. Possible mechanisms underlying slow component of $\dot{V}O_2$ on-kinetics in skeletal muscle. *J Appl Physiol (1985)*. 2015;118(10):1240–9.
27. Burnley M, Doust JH, Ball D, Jones AM. Effects of prior heavy exercise on $\dot{V}O(2)$ kinetics during heavy exercise are related to changes in muscle activity. *J Appl Physiol (1985)*. 2002;93(1):167–74.
28. Endo MY, Kobayakawa M, Kinugasa R, et al. Thigh muscle activation distribution and pulmonary $\dot{V}O_2$ kinetics during moderate, heavy, and very heavy intensity cycling exercise in humans. *Am J Physiol Regul Integr Comp Physiol*. 2007;293(2):R812–20.
29. Jones AM, Grassi B, Christensen PM, Krustrup P, Bangsbo J, Poole DC. Slow component of $\dot{V}O_2$ kinetics: mechanistic bases and practical applications. *Med Sci Sports Exerc*. 2011;43(11):2046–62.
30. Zoladz JA, Gladden LB, Hogan MC, Nieckarz Z, Grassi B. Progressive recruitment of muscle fibers is not necessary for the slow component of $\dot{V}O_2$ kinetics. *J Appl Physiol (1985)*. 2008;105(2):575–80.
31. Farina D. Interpretation of the surface electromyogram in dynamic contractions. *Exerc Sport Sci Rev*. 2006;34(3):121–7.
32. Lucía A, Hoyos J, Chicharro JL. The slow component of $\dot{V}O_2$ in professional cyclists. *Br J Sports Med*. 2000;34(5):367–74.
33. Cleuziou C, Perrey S, Borrani F, et al. $\dot{V}O_2$ and EMG activity kinetics during moderate and severe constant work rate exercise in trained cyclists. *Can J Appl Physiol*. 2004;29(6):758–72.
34. Colosio AL, Caen K, Bourgois JG, Boone J, Pogliaghi S. Metabolic instability vs fibre recruitment contribution to the $\dot{V}O_2$ slow component in different exercise intensity domains. *Pflugers Arch*. 2021;473(6):873–82.
35. Korzeniewski B, Rossiter HB. Exceeding a "critical" muscle P(i): implications for [formula: see text] and metabolite slow components, muscle fatigue and the power-duration relationship. *Eur J Appl Physiol*. 2020;120(7):1609–19.
36. Goulding RP, Rossiter HB, Marwood S, Ferguson C. Bioenergetic mechanisms linking $\dot{V}O_2$ kinetics and exercise tolerance. *Exerc Sport Sci Rev*. 2021;49(4):274–83.
37. Grassi B, Rossiter HB, Zoladz JA. Skeletal muscle fatigue and decreased efficiency: two sides of the same coin? *Exerc Sport Sci Rev*. 2015;43(2):75–83.
38. Ørtenblad N, Westerblad H, Nielsen J. Muscle glycogen stores and fatigue. *J Physiol*. 2013;591(18):4405–13.

APPLIED SCIENCES

39. Ørtenblad N, Nielsen J, Saltin B, Holmberg HC. Role of glycogen availability in sarcoplasmic reticulum Ca2+ kinetics in human skeletal muscle. *J Physiol*. 2011;589(Pt 3):711–25.
40. Cheng AJ, Chaillou T, Kamandulis S, et al. Carbohydrates do not accelerate force recovery after glycogen-depleting followed by high-intensity exercise in humans. *Scand J Med Sci Sports*. 2020;30(6):998–1007.
41. Lepers R, Maffiuletti NA, Rochette L, Brugniaux J, Millet GY. Neuromuscular fatigue during a long-duration cycling exercise. *J Appl Physiol (1985)*. 2002;92(4):1487–93.
42. Allen DG, Trajanovska S. The multiple roles of phosphate in muscle fatigue. *Front Physiol*. 2012;3:463.
43. Debold EP. Decreased myofilament calcium sensitivity plays a significant role in muscle fatigue. *Exerc Sport Sci Rev*. 2016;44(4):144–9.
44. Krüger RL, Aboodarda SJ, Jaimes LM, MacIntosh BR, Samozino P, Millet GY. Fatigue and recovery measured with dynamic properties versus isometric force: effects of exercise intensity. *J Exp Biol*. 2019;222(Pt 9):jeb197483.
45. Rossiter HB, Kowalchuk JM, Whipp BJ. A test to establish maximum O_2 uptake despite no plateau in the O_2 uptake response to ramp incremental exercise. *J Appl Physiol (1985)*. 2006;100(3):764–70.
46. Sundberg CW, Prost RW, Fitts RH, Hunter SK. Bioenergetic basis for the increased fatigability with ageing. *J Physiol*. 2019;597(19):4943–57.
47. Place N, Lepers R, Deley G, Millet GY. Time course of neuromuscular alterations during a prolonged running exercise. *Med Sci Sports Exerc*. 2004;36(8):1347–56.
48. Goodall S, Thomas K, Harper LD, et al. The assessment of neuromuscular fatigue during 120 min of simulated soccer exercise. *Eur J Appl Physiol*. 2017;117(4):687–97.
49. Brownstein CG, Metra M, Pastor FS, Faricier R, Millet GY. Disparate mechanisms of fatigability in response to prolonged running versus cycling of matched intensity and duration. *Med Sci Sports Exerc*. 2022;54:872–82.
50. Finn HT, Rouffet DM, Kennedy DS, Green S, Taylor JL. Motoneuron excitability of the quadriceps decreases during a fatiguing submaximal isometric contraction. *J Appl Physiol (1985)*. 2018;124(4):970–9.
51. Brownstein CG, Espeit L, Royer N, et al. Reductions in motoneuron excitability during sustained isometric contractions are dependent on stimulus and contraction intensity. *J Neurophysiol*. 2021;125(5):1636–46.
52. McNeil CJ, Giesebrecht S, Gandevia SC, Taylor JL. Behaviour of the motoneurone pool in a fatiguing submaximal contraction. *J Physiol*. 2011;589(Pt 14):3533–44.
53. Johnson KV, Edwards SC, Van Tongeren C, Bawa P. Properties of human motor units after prolonged activity at a constant firing rate. *Exp Brain Res*. 2004;154(4):479–87.
54. Krüger RL, Aboodarda SJ, Jaimes LM, Samozino P, Millet GY. Cycling performed on an innovative ergometer at different intensities-durations in men: neuromuscular fatigue and recovery kinetics. *Appl Physiol Nutr Metab*. 2019;44(12):1320–8.
55. Fullerton MM, Passfield L, MacInnis MJ, Iannetta D, Murias JM. Prior exercise impairs subsequent performance in an intensity- and duration-dependent manner. *Appl Physiol Nutr Metab*. 2021;46(8):976–85.
56. Clark IE, Vanhatalo A, Thompson C, et al. Dynamics of the power-duration relationship during prolonged endurance exercise and influence of carbohydrate ingestion. *J Appl Physiol (1985)*. 2019;127(3):726–36.
57. Miura A, Sato H, Sato H, Whipp BJ, Fukuba Y. The effect of glycogen depletion on the curvature constant parameter of the power-duration curve for cycle ergometry. *Ergonomics*. 2000;43(1):133–41.
58. Gathercole R, Sporer B, Stellingwerff T, Sleivert G. Alternative countermovement-jump analysis to quantify acute neuromuscular fatigue. *Int J Sports Physiol Perform*. 2015;10(1):84–92.

APPLIED SCIENCES